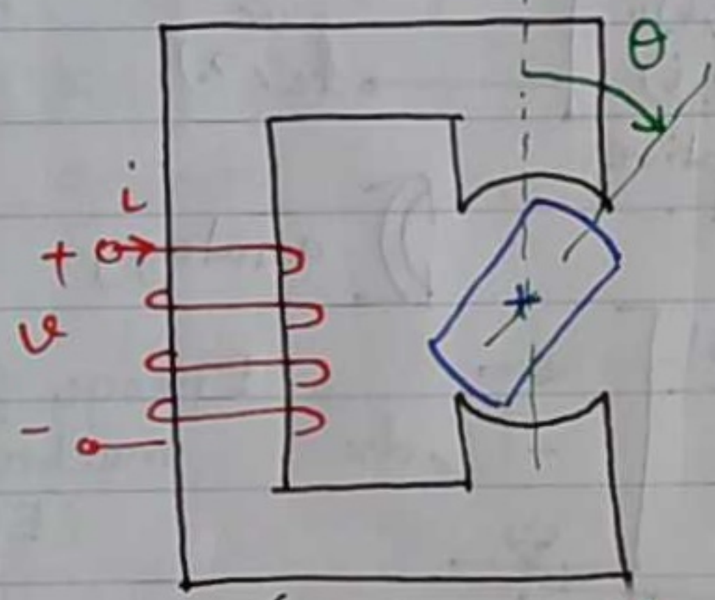
Lecture 25.

Last time : Q. How can we produce continuous motion ? Cyclic operation.

- Excite windings/coils based on a position feedback.
- Please sketch the operation in different planes.

control → (θ, i)

(i, λ) plane, (θ, τ_e) plane, (θ, λ) plane.

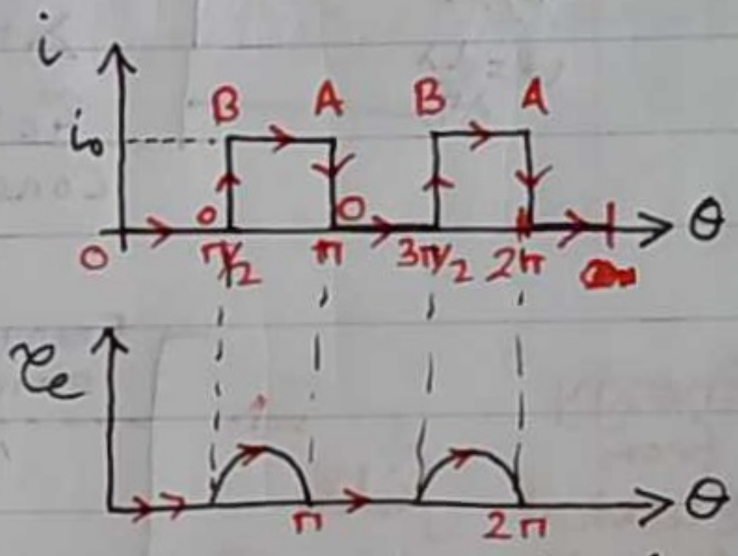


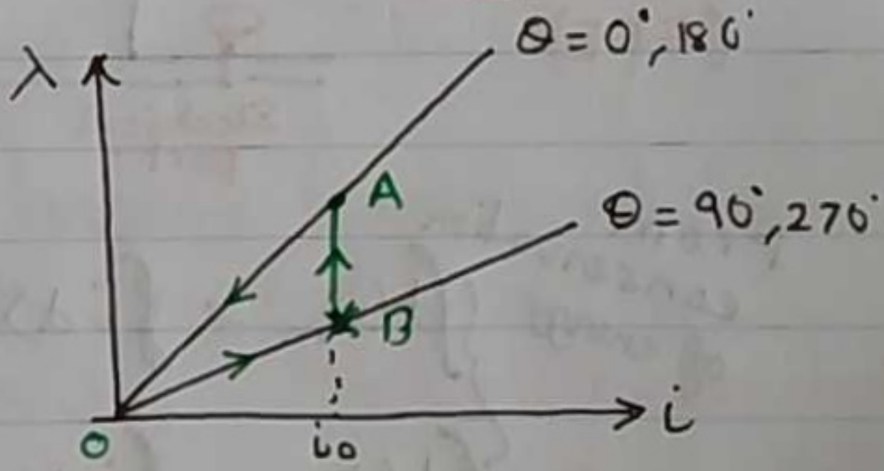
$$L(\theta) = (L_0 + L_1 \cos 2\theta)$$

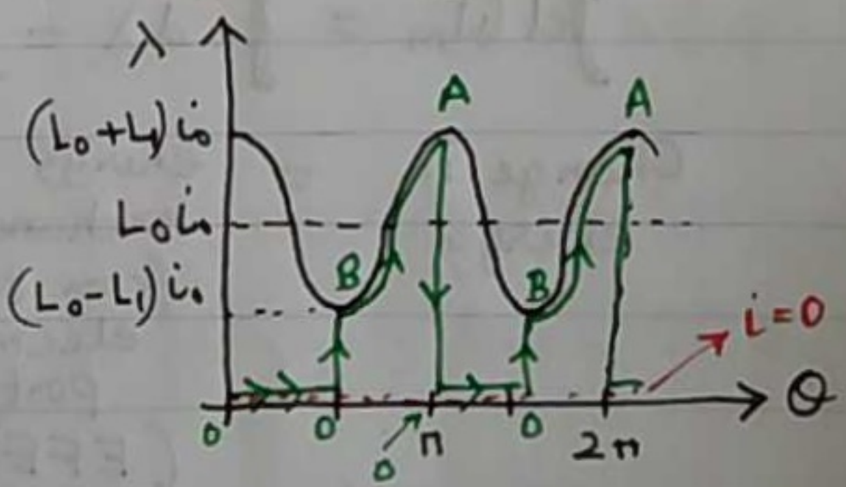
$$\lambda = L(\theta) \cdot i$$

$$\omega_m' = \frac{1}{2} L(\theta) i^2$$

$$\tau_e = -i^2 L_1 \sin 2\theta .$$

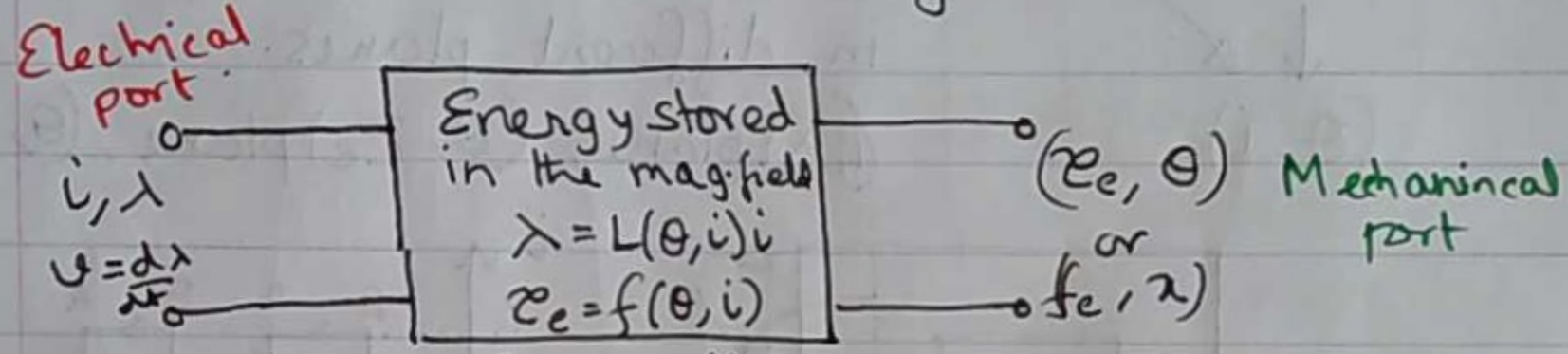




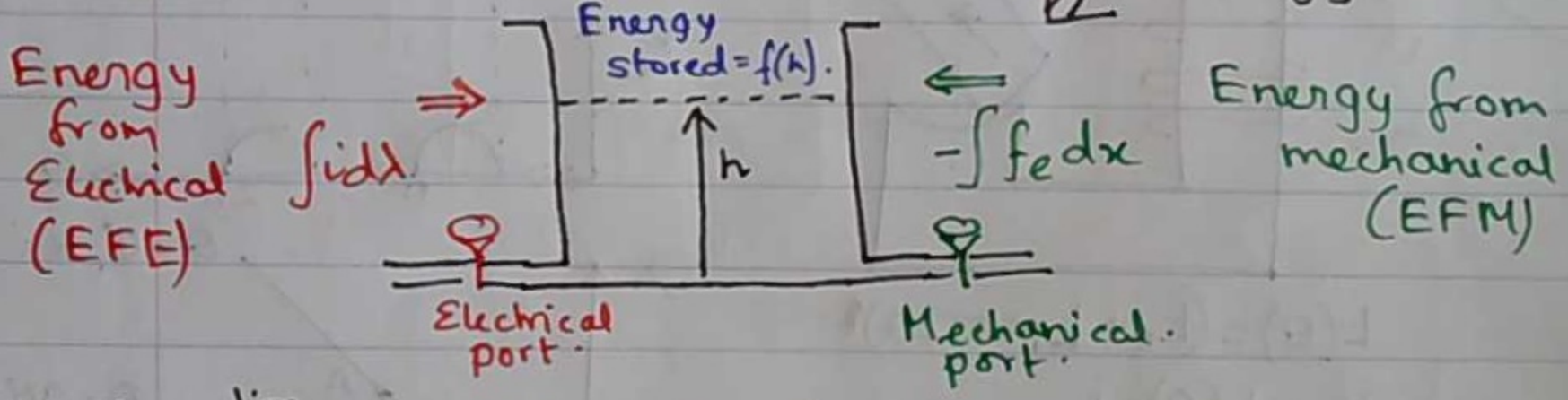


Today: Q. How can we use graphical analysis to understand energy conversion?

Q. How can we identify the direction of energy flow? — Motor or generator?



Conservative field.



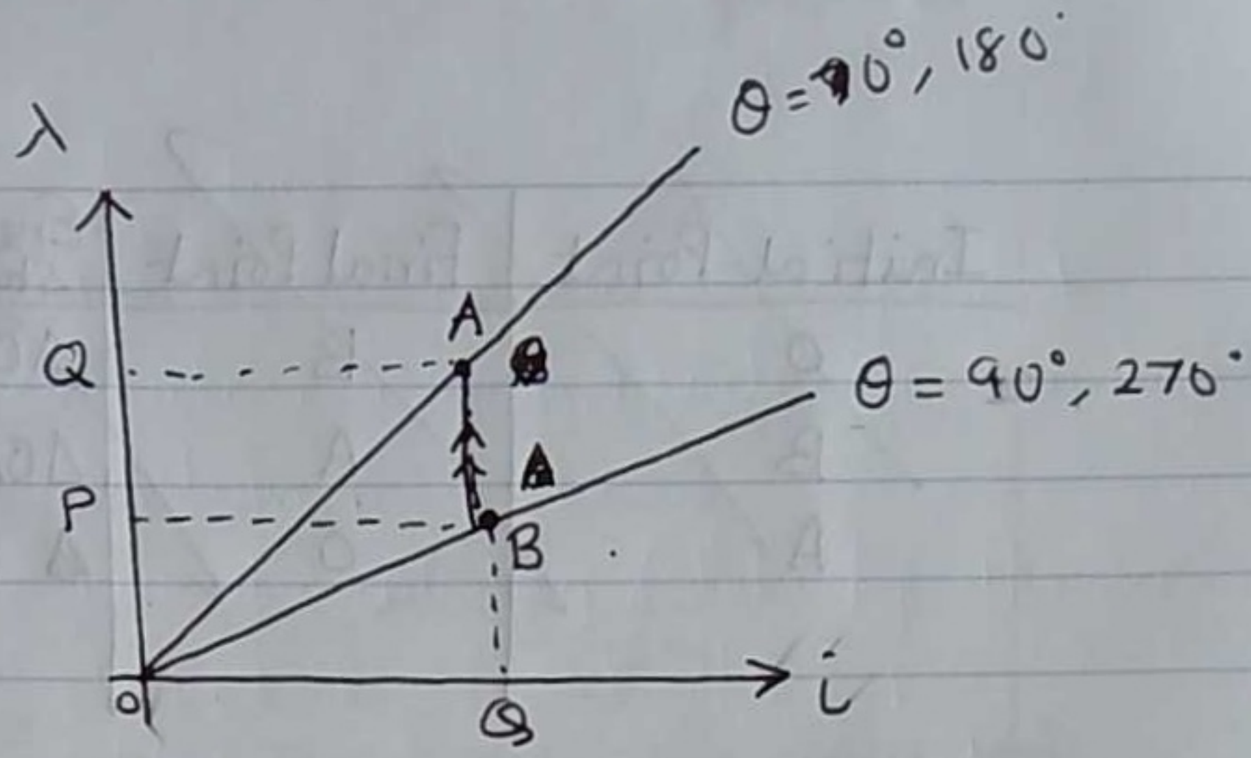
analogy.

From conservation of energy

$$\int dW_m = \int i\,d\lambda - \int f_e\,dx \Leftarrow \text{translational system.}$$

$$\int dW_m = \int i\,d\lambda - \int \tau_e\,d\theta \Leftarrow \text{rotational system}$$

Change in energy = energy exchanged from the electrical port. (EFE) + energy exchanged from the mechanical port. (EFM).



Energy stored in the magnetic field at each $(\vec{\lambda}, i, \theta)$ point in uniquely defined, independent of path, because the field is conservative. Similar to the gravity field, (h).

Eg: $W_m\big|_A$ = Area OAP ; $W_m\big|_B$ = Area OBQ.

why? $W_m = \int_{@\,\theta\,\text{const}} i\,d\lambda$

Q. Can we calculate energy transferred from the electrical port to the field as we move from B to A? Mind it!! $\theta \neq$ constant

$$EFE_{B\to A} = \int_{B}^{A} i\,d\lambda = \text{Area } PABQ.$$

This is dependent on the path.

What about: $EFE_{O\to B} = \int_{O}^{B} i\,d\lambda$ = Area OBQ.

Initial Point	Final Point	Energy stored	EFE
O	B	$\triangle OBP$	$\triangle OBP$
B	A	$\triangle OAQ$	$\square PBAQ$
A	O	0	$-\triangle OAQ$

$\triangle OAQ$

Initial Point	Final Point	Energy stored at the final point	EFE (dependent on trajectory	EFM $= \Delta W -$
O	B	$\triangle OBP$	$\triangle OBP$	0 mal sen $\theta =$
B	A	$\triangle OAQ$.	$\square PBAQ$.	$-\triangle OAB$ indica
A	O	0	$-\triangle OAQ$	0 fie

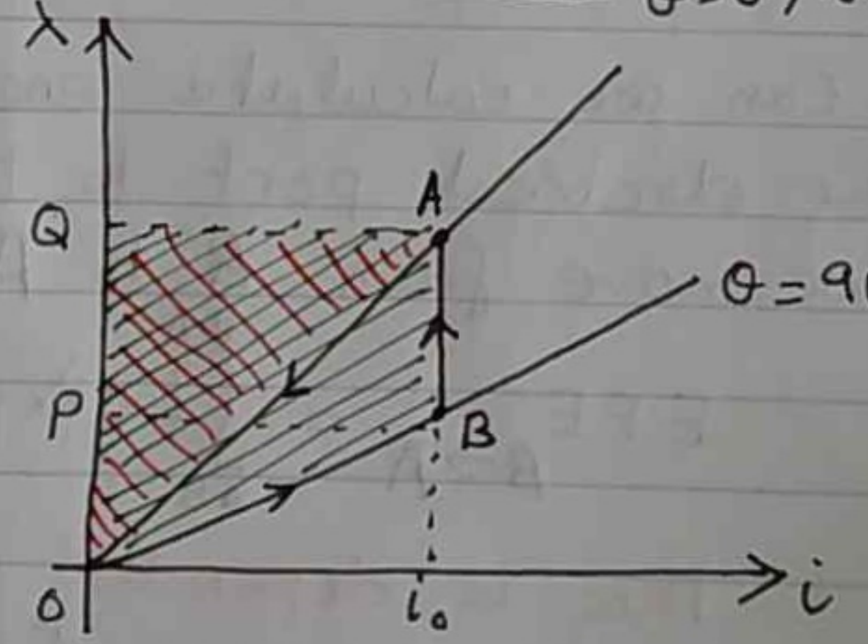
$\theta = 0°, 180$

$$\int_B^A d\omega_m = \int_B^A i\,d\lambda - \int_B^A \tau_e\,d\theta$$

$$\int_B^A d\omega_m - \int_B^A i\,d\lambda$$

$$-\int_B^A \tau_e\,d\theta = \omega_m|_A - \omega_m|_B - \int_B^A i\,d\lambda$$

EFM



$$= \triangle OAQ - \triangle OPB - \square PBAQ$$

$$= -\triangle OAB.$$

* $d\omega_m = i\,d\lambda - f_e\,dx$

* $\int_A^B d\omega_m = \underbrace{\int_A^B i\,d\lambda}_{\text{EFE}} - \underbrace{\int_A^B f_e\,dx}_{\text{EFM}}$

$\int_A^B d\omega_m$ — independent of path

EFE, EFM — These are path dependent.

* $0 = \oint d\omega_m = \oint_C EFE + \oint_C EFM$

* If EFE > 0, it is a motor over a cycle. i.e over a cycle, energy flows from elec. port to mech. port.

* If EFE > 0, EFM < 0 over a cycle.

* If EFE < 0, it is a generator over a cycle.